

Bernoulli's Equation: $y' + p(x)y = q(x)y^n$

Derivation:

Step 1

$$y' + p(x)y = q(x)y^n$$

Divided by y^n :

$$y^{-n}y' + p(x)y^{1-n} = q(x)$$

Let $z = y^{1-n}$, $z' = (1-n)y^{-n}y'$, i.e. $y^{-n}y' = (1-n)^{-1}z'$

$$y^{-n}y' + p(x)y^{1-n} = q(x)$$

$$(1-n)^{-1}z' + p(x)z = q(x)$$

Simplifying:

$$z' + (1-n)p(x)z = (1-n)q(x).$$

Step 2 Solve the first order linear differential equation using the integrating factor method.

Bernoulli's Equation Solution Steps:

$$y' + p(x)y = q(x)y^n$$

Step 1 Find out n

Step 2 Set $z = y^{1-n}$

Step 3 Directly transform to

$$z' + (1-n)p(x)z = (1-n)q(x).$$

Step 4 Solve the DE using the integrating factor method

Homogenous 2nd Order Linear ODEs with Constant Real Coefficients:

$$y'' + ay' + by = 0, \quad a, b \in \mathbb{R}$$

Step 1 Find out the **characteristic equation** $\lambda^2 + a\lambda + b = 0$. Then solve for λ .

$$\lambda_1, \lambda_2 = \frac{1}{2}(-a \pm \sqrt{a^2 - 4b})$$

Step 2 Write down the particular solutions, and corresponding general solution:

<u>Case A:</u>	<u>Case B:</u>	<u>Case C:</u> λ_1, λ_2 complex:
λ_1, λ_2 real and distinct	λ real, repeated	$\lambda_1, \lambda_2 = \alpha \pm \beta i$
$y_1 = e^{\lambda_1 x}, y_2 = e^{\lambda_2 x}$	$y_1 = e^{\lambda x}, y_2 = xe^{\lambda x}$	$y_1 = e^{\alpha x} \cos \beta x, y_2 = e^{\alpha x} \sin \beta x$
$y = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$	$y = C_1 e^{\lambda x} + C_2 x e^{\lambda x}$	$y = C_1 e^{\alpha x} \cos \beta x + C_2 e^{\alpha x} \sin \beta x$

Non-homogeneous 2nd Order ODE:

$$y'' + p(x)y' + q(x)y = r(x)$$

- If $r(x) \neq 0$, the DE becomes **non-homogeneous**. The general solution is:

$$y = y_h + y_p = (C_1 y_1 + C_2 y_2) + y_p$$

y_h : Solution to the corresponding homogeneous ODE

$$\text{i.e. } y'' + p(x)y' + q(x)y = 0$$

y_p : a particular solution to the original non – homogeneous ODE

$$\text{i.e. } y'' + p(x)y' + q(x)y = r(x)$$

• **To Find y_p : Two methods**

- i) *Method of Undetermined Coefficients*
- ii) *Method of Variation of Parameter*

➤ **Method of Undetermined Coefficients:**

$$y'' + p(x)y' + q(x)y = r(x), \quad r(x) \neq 0$$

Categorize based on $r(x)$

Case A: polynomial	Case B: exponential $r(x) = P(x)e^{kx}$	Case C: trigonometric $r(x) \equiv P(x)e^{\alpha x} \sin \beta x$ or $r(x) \equiv P(x)e^{\alpha x} \cos \beta x$
Guess y_p to be a <u>polynomial with unknown constant coefficients with the same highest power</u> as the highest order of $r(x)$. Otherwise, choose higher powers.	Guess $y_p = u(x)e^{kx}$, where u is a polynomial.	Guess $y_p = ue^{(\alpha+i\beta)x}$. ➤ If $r(x)$ has $\sin \beta x$, then $y_p = \text{Im } ue^{(\alpha+i\beta)x}$. ➤ If $r(x)$ has $\cos \beta x$, then $y_p = \text{Re } ue^{(\alpha+i\beta)x}$.

Remarks

Choose higher power polynomial if your choice doesn't work well, to be specific:

- A. If the value of k is **neither** λ_1 or λ_2 , then guess u to be a polynomial with the highest power same as $P(x)$.
- B. If the value of k tallies with **either** simple root λ_1 or λ_2 , then guess u to be a polynomial with highest power + 1 of highest power of $P(x)$.
- C. If the value of k tallies with repeated root λ , then guess u to be a polynomial with highest power + 2 of highest power of $P(x)$.

➤ **Method of Variation of Parameters:** $y'' + p(x)y' + q(x)y = r(x), \quad r(x) \neq 0$

Step 1 Find Wronskian $W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_1' y_2$.

Step 2 Use the formula to find u and v .

$$u = - \int \frac{y_2 r}{W(y_1, y_2)} dx \qquad v = \int \frac{y_1 r}{W(y_1, y_2)} dx$$

Step 3 $y_p = uy_1 + vy_2$

SUMMARY OF 2ND ORDER ODE:

$$y'' + p(x)y' + q(x)y = r(x)$$

- If $r(x) \equiv 0$, this is known as a **homogeneous DE**.

Step 1 Solve the characteristic equation $\lambda^2 + a\lambda + b = 0$ for λ_1, λ_2 .

Step 2 Find whether the roots are in Case A, B or C. The general solution is:

$$y = C_1y_1 + C_2y_2$$

- If $r(x) \neq 0$, the DE becomes **non-homogeneous**.

The general solution is:

$$y = y_h + y_p = (C_1y_1 + C_2y_2) + y_p$$

y_h : *Solution to the corresponding homogeneous ODE*

$$\text{i.e. } y'' + p(x)y' + q(x)y = 0$$

y_p : *a particular solution to the original non – homogeneous ODE*

$$\text{i.e. } y'' + p(x)y' + q(x)y = r(x)$$

To Find y_p :

iii) *Method of Undetermined Coefficients – polynomial, exponential, sin/cos*

iv) *Method of Variation of Parameter*